

C $\nearrow$ Linear
generic
trust game $\checkmark$.

$$u_1(T, C) = 25.$$

$$u_2(T, C) = 23.$$

$$u(T, D) = (-100, 150).$$

Normal form game (NFG) 无先占.
 $\int \rightarrow ? \text{ package ? } ?$

$$\Delta(X) = \{ (p_1, \dots, p_n) : p_i \geq 0, \sum_{i=1}^n p_i = 1 \}$$

$$\Delta([c, D]) = \{ (p, 1-p) : 0 \leq p \leq 1 \}.$$

line.

$$A_1^{\text{sim}} = \{T, W, S\}.$$

$$\Delta(\{T, W, S\}) = \{ \pi_i = (P_t, P_w, P_s), P_t, P_w, P_s \geq 0, P_t + P_w + P_s = 1 \}$$

Triangle.

$$\psi: \pi: A_2 \rightarrow \Delta(A_1)$$

$$\pi = (\pi_1, \pi_2)$$

$$\pi_1 \in \Delta(A_1), \pi_2 \in \Delta(A_2)$$

Strategy
policy.

$$\Pi = \Pi_1 \times \Pi_2 = \Delta(A_1) \times \Delta(A_2)$$

$$P(a, b) = \pi_1(a) \cdot \pi_2(b)$$

$$\text{Supp}(\pi_i) = \{a \in A_i : \pi_i(a) > 0\}.$$

$$|\text{Supp}(\pi_i)| = 1 \rightarrow \text{pure}.$$

$$u_i(\pi) = \sum_{(a,b) \in A} \pi_1(a) \pi_2(b) u_i(a,b)$$

$$u_1(\pi) = 0.2 \times (25) + 0.05(-100) + 0.6(0) + 0.15(0) \\ = 0$$

$$u_2(\pi) = 0.2(25) + 0.05(150) + 0.6(0) + 0.15(0) \\ = 12.5$$

$$\pi_1 \in \arg \max_{\pi'_1 \in \Pi_1} u_1(\pi'_1, \pi_2)$$

best response: $\pi_1 \rightarrow \pi_2$.

VS

$$\max_{\pi'_1} u_1 \rightarrow \underline{u_1}$$

$br(\pi_2)$: pure best responses.

indifference.

$$u_1(br, \pi_2) := \max_{a \in A_1} u_1(a, \pi_2)$$

Nash equilibrium. $\neq$ best result.

$$NECG := \begin{cases} u_1(\pi_1, \pi_2) \geq u_1(\pi'_1, \pi_2) \quad \forall \pi'_1 \in \Pi_1 \\ u_2(\pi_1, \pi_2) \geq u_2(\pi_1, \pi'_2) \quad \forall \pi'_2 \in \Pi_2 \end{cases}$$

at same time.

Profit deviation

• Pure + commitment equilibrium. Stackelberg.

有先后.

Pure
P₂ commit Action b
(leader) ↓

P₁ observe b

P₁ choose best response. a.

U₁(a, b), U₂(a, b).

SE P²_{pure}(G)
Stackelberg equilibrium policy.

① $\psi_{br}(b) \in \Delta(b|b)$

② $b \in \arg \max_{b' \in A_2} E_{a \sim \psi_{br}(b')} U_2(a, b')$

Bob commit C → U₁(T, C) = 25 ✓ → T → U₂ = (T, C) = 25.

U₁(W0, C) = 0 X

Bob commit D → U₁(T, D) = -100 X

U₁(W0, D) = 0 ✓ → W0 → U₂ = (W0, D) = 0.

Optimal commitment is C
now NE(G) is (T, C)

but in ~~non~~ NE(G), NE(G) is (W0, D)

• Best response Policy $\psi_{br}: b' \in A_2 \mapsto \psi_{br}(b') \in \Delta(b|b')$

$\psi_{br}(C) = T$ $\psi_{br}(D) = W0$.

$\psi_{br} = \{ C \mapsto T, D \mapsto W0 \}$.

$$b' = C$$

$$\psi_{br}(C) = T$$

$$u_2(T, C) = 25$$

$$b' = D$$

$$\psi_{br}(D) = W0.$$

$$u_2(W0, D) = 0.$$

$$C \in \arg \max \{25, 0\} \xrightarrow{\text{output}} (T, C)$$

$$C \psi_{br, C} = (T, C) \in SEP_{\text{pure}}^2(G).$$

tie-breaking ? $a, a' \in br(b)$

• generic-pay off games:

why?

$$u_i(a, b) \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1]$$

generic-pay off games G no best-response utility tiebreaking.

$$NFG: G = (A, u)$$

$$\text{Mixed strategies } \pi_1, \pi_2$$

$$\text{Expected utilities } u_i(\pi)$$

$$\text{Best Responses } br(\pi')$$

$$NE(G)$$

P_2 commit first
then P_1

Pure-commitment

→ No utility tiebreaking.
Generic
Payoffs

3. P_1 : simulator G G_{sim}^C

P_2 : simulated player

BR policy: $br(b) \rightarrow \text{utilitarianism}$

$$\Psi_{br} := \left\{ \psi_{br} : A_2 \rightarrow \Delta(A_1) \mid \forall b \in A_2 : \psi_{br}(b) \in \Delta(br(b)) \right\}$$

$\uparrow$ action of
input P_2

$$\begin{cases} b \in A_2 \\ \psi_{br}(b) \in \Delta(A_1) \\ \psi_{br}(b) \in \Delta(br(b)) \end{cases}$$

$$\Psi_{br}^{det} \subset \Psi_{br} \quad |\Psi_{br}^{det}| = 1$$

one deterministic pure best response for b .

D4: (1) $G_{sim}^{C, all}$

$$\psi_{br} \in \Psi_{br}^{det} \text{ s.t. } u_1(s_{\psi_{br}}, b) := u_1(\psi_{br}(b), b) - c$$

$$u_2(s, b) := u_2(\psi_{br}(b), b)$$

(2)

Probably different for u_2

$$u_2(s, b) = \mathbb{E}_{a \sim \psi_{br}(b)} u_2(a, b)$$

G_{sim}^C

Lemma 5 $\forall b \in A_2, a, a' \in br(b) \Rightarrow u_2(a, b) = u_2(a', b)$.

Lemma 6: $NE(G_{sim}^{c, all}) \setminus NE(G)$

$$= \bigcup_{b \in br} [NE(G_{sim}^{c, all}) \setminus NE(G)].$$

P7:

(i) $c < 0, u_i(s, b) = u_i(br, b) - c$

dominant

$u_i(s, b) > u_i(br, b) \geq u_i(a, b) \quad \forall a \in A_1$
pure-commit game.

$$NG(G_{sim}^c) \subseteq SEP_{pure}^2(G) \quad c < 0$$

(ii) c large

$$c > \max_{a, b} u_i(a, b) - \max_{\pi_1} \min_{\pi_2} u_i(\pi_1, \pi_2)$$

dominated

Security value.

$$NG(G_{sim}^c) = NE(G)$$

D8: $V_{0IS}(\pi_2) := \left(\sum_{b \in A_2} \pi_2(b) \max_{a \in A_1} u_i(a, b) \right) - \max_{\pi_1 \in \Pi_1} u_i(\pi_1, \pi_2)$

~~expect~~ max expect?



$$= \left(\sum_{b \in A_2} \pi_2(b) \max_{a \in A_1} u_i(a, b) \right) - \left(\max_a \sum_b \pi_2(b) u_i(a, b) \right)$$

TG: $V_0 I_S$

let $\text{bob}^P(\text{defect}) : q = P(D)$

$$P(C) = 1 - q$$

if $C=0 : (1-q) 25 + q(0) = 25(1-q)$

If no accurate info of (simulates)

trust: $u_1(T, \pi_2) = 25(1-q) - 100q = 25 - 125q$

WO: $u_1(WO, \pi_2) = 0$

$\max \{ 25 - 125q, 0 \}$

$\therefore V_0 I_S(q) = (25 - 25q) - \max \{ 25 - 125q, 0 \}$

$25 - 125q = 0$
 $q = 0.2$

i) $0 \leq q \leq 0.2$ $V_0 I_S(q) = 25 - 25q - (25 - 125q) = 100q$

ii) $q > 0.2$ $V_0 I_S(q) = 25 - 25q$

$(0.2 < q \leq 1)$

Lemma 9:
$$u_1(S, \pi_2) = \overbrace{u_1(b_r, \pi_2)}^{\text{sim}} + \overbrace{V_0 I_S(\pi_2)}^{\text{No-sim}} - C$$

$$= u_1(b_r, \pi_2) + \sum_b \pi_2(b) \max_a u_1(a, b) - u_1(b_r, \pi_2) - C$$

$$= \sum_b \pi_2(b) \max_a u_1(a, b) - C$$

Lemma 10: If $0 < \pi_1(S) < 1$, $V_0 I_S(\pi_2) = C$.

No difference between S or NOT S in this case

4: Structural Properties:

$$-\infty = e_{-1} < 0 = e_0 < e_1 < \dots < e_k < e_{k+1} = \infty$$

$$\forall c_0 \in (e_i, e_{i+1}) \text{ and } \pi^{c_0} \in NE(G_{sim}^{c_0}), \quad t_2: c \in [e_i, e_{i+1}] \mapsto \pi_2^c \in \Pi_2$$

$$\text{s.t. } t_2(c) = \pi_2^{c_0} \text{ and } (\pi_1^{c_0}, t_2(c)) \in NE(G_{sim}^c)$$

$$0 < c < 20 \quad \text{Let } A: P(T) = x, P(S) = z: \quad x + z = 1$$

$$\text{Bob: } P(D) = v \dots \{0, 25 - 2c\}$$

$$\text{Bob: } U_B(D) = U_B(c)$$

$$150x + 0 \cdot z = 25(x + z) = 25$$

$$x = \frac{1}{6} \quad z = \frac{5}{6}$$

$$\boxed{P_A(T) = \frac{1}{6} \quad P_B(S) = \frac{5}{6}}$$

~~X~~

Simulator strategy is piecewise constant.

$$\text{Alice: } U_A(T) = U_B(S)$$

$$25 - 125q = \cancel{U_A(S)} (25 - c)(1 - q) + (c - c)q$$

$$25 - 125q = 25 - 25q - c$$

$$q = \frac{c}{100}$$

$$\boxed{P_B(D) = \frac{c}{100} \\ P_B(c) = 1 - \frac{c}{100}}$$

No difference between 2 or not in the case